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import os
import sys

sys.path.insert(0,
os.path.abspath(os.path.join(os.path.dirname(__f
ile__), "..")))

from phishing_detector.extractor import
parse_message
from phishing_detector.heuristics import
run_heuristics

def test_detects_urgency_language():
    findings =
run_heuristics(parse_message("Please act now,
your account has been locked."))
    rules = [f.rule for f in findings]
    assert "urgency_language" in rules

def test_detects_ip_based_url():
    findings =
run_heuristics(parse_message("Click here:
http://192.168.1.50/login"))
    rules = [f.rule for f in findings]
    assert "ip_based_url" in rules

def test_detects_shortener():
    findings =
run_heuristics(parse_message("Check this out:
http://bit.ly/abc123"))
    rules = [f.rule for f in findings]
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    assert "url_shortener" in rules

def test_clean_message_has_no_findings():
    findings =
run_heuristics(parse_message("Hey, lunch
tomorrow at noon?"))
    assert len(findings) == 0

def test_detects_lookalike_domain():
    findings =
run_heuristics(parse_message("Verify here:
http://paypal-secure-login.xyz/verify"))
    rules = [f.rule for f in findings]
    assert "lookalike_domain" in rules

def test_detects_sender_brand_mismatch():
    raw = (
        "From: PayPal Security
<security@totally-not-paypal.top>\n"
        "Subject: Account alert\n\n"
        "Please review your account.\n"
    )
    findings =
run_heuristics(parse_message(raw))
    rules = [f.rule for f in findings]
    assert "sender_brand_mismatch" in rules

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